Example longform document

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Preface

This example demonstrates the longform template: front matter, chapters, sections, theorem-like environments, cross-references, citations, and a bibliography.

CHAPTER 1

First chapter

1. Basic structure

This is ordinary mathematical text in a chaptered document. We cite [Har77] and refer to Theorem 1.2.

Definition 1.1. A *sample object* is an object introduced to demonstrate the formatting of long-form mathematical writing.

THEOREM 1.2. *Let X be a sample object. Then X satisfies the following properties:*

(i) X has a first property.
(ii) X has a second property.

PROOF. This is a sample proof. The displayed equation

$$\chi(\mathcal{F}) = \sum_i (-1)^i h^i(X, \mathcal{F})$$

checks the display style in a long-form document. □

The first property is Item (i).

Proposition 1.3. *The sample object of Definition 1.1 also satisfies a proposition.*

Remark 1.4. This is a remark in the longform template.

CHAPTER 2

Second chapter

1. A diagram

The template supports commutative diagrams:

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D. \end{array}$$

2. References inside the document

We have used Definition 1.1, theorem 1.2, and proposition 1.3 and the citation [Har77].

Bibliography

[Har77] Robin Hartshorne. *Algebraic Geometry*. Vol. 52. Graduate Texts in Mathematics. Springer, 1977.